

Orion Robotics

Orion A1 – Vision Module Reference

© 2025 Orion Robotics – Vision Systems Engineering Division

The Orion A1 vision module enables spatial awareness and visual understanding across a range of dynamic environments. It features dual 120° stereo cameras mounted within the head assembly, providing depth perception and object tracking capabilities. This subsystem operates at 60 frames per second, continuously fusing imagery from visible and infrared bands to ensure reliable navigation, obstacle avoidance, and human interaction.

Each camera assembly is independently adjustable to correct for manufacturing tolerances, ensuring sub-millimetre alignment precision. The system uses depth triangulation to generate a real-time 3D model of its surroundings, which is then passed to the motion controller for situational adaptation.

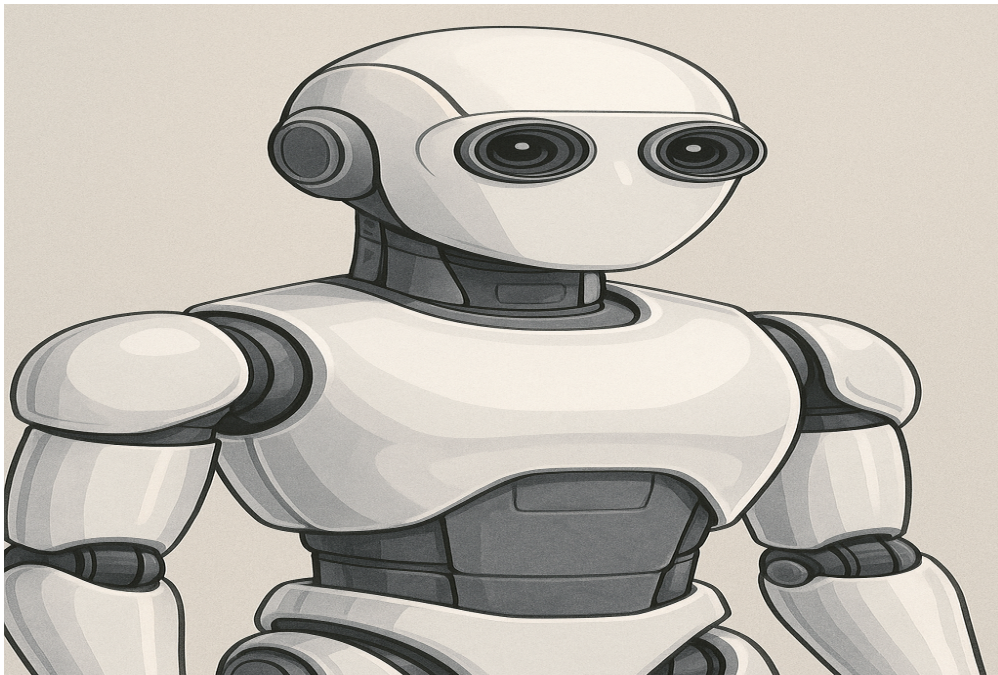


Figure 1. Orion A1 humanoid vision module with dual stereo cameras.

1. Calibration and Alignment

To achieve high-precision depth mapping, the Orion A1 vision module requires regular calibration. Technicians initiate this process using the Orion Vision Calibration Tool (OVCT), which employs checkerboard patterns to correct lens distortion and compute stereo alignment parameters. Calibration should be performed in controlled lighting conditions to minimize reflection artifacts that could mislead the disparity map generator.

Once calibration is complete, the firmware stores a unique intrinsics profile for each camera. Any deviation exceeding 0.02 degrees between optical axes triggers a maintenance alert. During autonomous operation, the robot uses visual landmarks to continuously self-correct its orientation when GPS or motion data is unavailable.

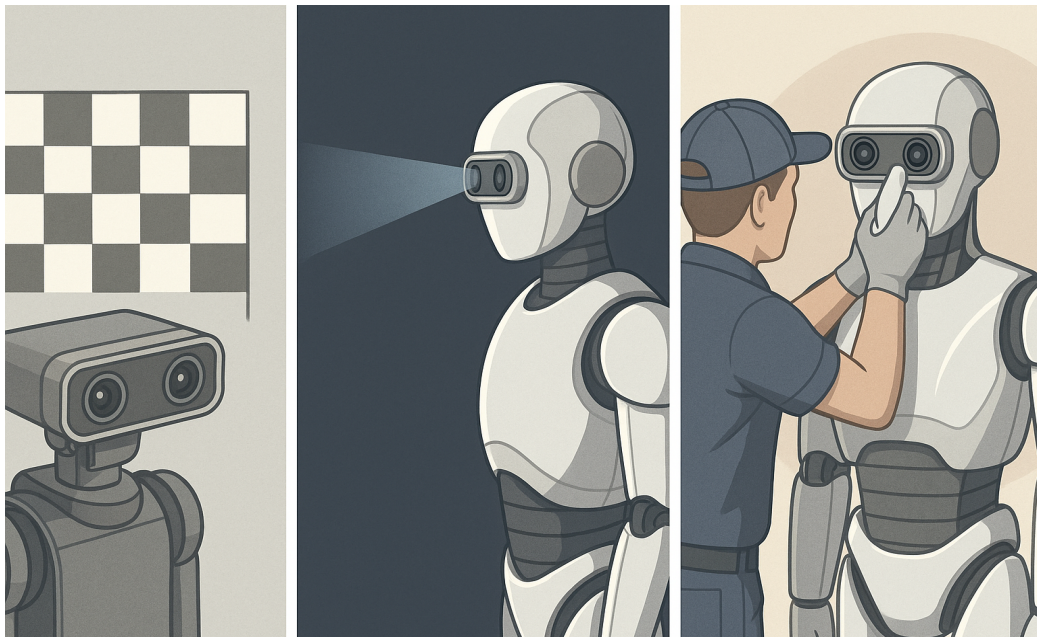


Figure 2. Left: Calibration using checkerboard pattern. Center: Low-light depth projection. Right: Field technician performing lens maintenance.

2. Operation and Maintenance

In field conditions, the Orion vision module must operate across lighting extremes—from dim warehouses to bright outdoor sunlight. An adaptive exposure algorithm dynamically adjusts gain and shutter parameters to prevent saturation while preserving image contrast.

Infrared-assisted depth sensing automatically activates under lux levels below 40, ensuring continuity of perception.

Maintenance procedures recommend cleaning the optical lenses every 250 operational hours using the Orion LensCare Kit. Engineers must also inspect the internal gimbal assembly for dust accumulation that can affect autofocus. All lens replacements require re-running OVCT to maintain stereo calibration integrity.

Key Takeaways:

- Regular calibration ensures sub-millimetre alignment precision.
- The vision system adapts dynamically to lighting conditions.
- Maintenance intervals should not exceed 250 operational hours.
- Infrared-assisted depth sensing maintains perception in darkness.

End of Document